Response to reviewers:

We would like to explicitly point out that this is the first work to report the universal SOI effect in Mie scattering, and also construct one microscopic electrodynamic model to bridge the near-field SOI and the polarized lights in far fields, through strictly deriving the continuity equation of photonic spin and orbital angular momenta in an inhomogeneous medium. The reported phenomenon and established model own broad influences in optics. The requested clarifications, additional work, and lists of changes are now provided, as discussed below.

Referee 1:

1. My primary concern is the novelty of the study. The spin Hall effect in plasmonic nanoparticles has been extensively theoretically analyzed and experimentally demonstrated in prior works, such as [PRL 102, 123903 (2009)][PRL 104, 253601, 2010], [PRL 117, 166803, 2016]. The authors need to clearly differentiate their work from previous studies to justify the novelty of their current research. Specifically:

a. Deng Pan et al. (PRL 117, 166803, 2016) have studied spin-orbit interaction in 80 nm gold nanoparticles and provided theoretical analysis of the phenomenon, which appears quite similar to the system studied in this manuscript. The authors need to provide a clear differentiation of their work from this study.

b. Experimentally, O. Rodrıguez-Herrera et al. (PRL 104, 253601, 2010) have also demonstrated strong spin-orbit interaction of light from 80 nm gold nanoparticles. The authors differentiate their work by stating that the previous demonstration of photonic SHE require nonparaxial conditions. However, the current work also utilizes a relatively large NA objective (0.8), which is difficult to consider as paraxial. Do the authors claim to observe photonic SHE even with parallel light incidence?

> Thanks for this constructive comment. The SOI phenomenon reported in this work is a new and universal effect in Mie scattering, and always accompanies the Mie scattering process. This phenomenon is one important property of Mie scattering that has not been reported in other works. Considering the extensive influence of Mie scattering, this SOI phenomenon will also find broad applications in diverse fields, including atmospheric science, remote sensing, biomedical optics, nanophotonics, environmental monitoring, astronomy, materials characterization, and beyonds. Thus, we revise the whole manuscript, including the name, to highlight the Mie scattering nature.

In addition, we construct one complete microscopic electromagnetic dynamics model to clearly reveal the whole SOI process in Mie scattering for the first time. Based on the rigorously-derived continuity equation of photonic spin and orbital angular momenta in an inhomogeneous medium, we clearly display the excitation process with polarized light, the intense SOI in near-field, and the polarized emission in far-field, where the spin momentum, orbit momentum, and optical force are involved. Obviously, this model can apply to all of the SOI phenomenon within the framework of Maxwell's theory, and greatly promote people's understanding of SOI effects.

The key difference with prior works is that the SOI phenomenon reported in this work is one fundamental property of Mie scattering, which is universal and own important influence in Mie region. In sharp contrast, the prior works, however, can't be summarized as one universal property of Mie scattering. For examples,

Deng Pan et al. (*PRL 117, 166803, 2016*) report that the SAM in excitation field can transfer to the twist streamlines near NP (i.e., orbital momentum), which, however, localizes in the near-field, and will vanish when the distance from NP is larger than several wavelengths. This twist streamline in near-field can't propagate to a long-distance unless the wave-vector matching condition is satisfied, for example, putting the NP on a substrate or gold nanowire, where a directional transmission can be observed [**]. Thus, this is a near-field and localized SOI effect, and is inherently different with the SOI effect in Mie scattering;

D. Haefner et al. (*PRL 102, 123903 (2009)*) report a far-field SOI effect when exciting NP with circularly polarized lights, where the far-field Poynting vector S will make an angle with the line of direct sight. But the authors also emphasized that this phenomenon will vanish under the excitation of linearly polarized lights;

O. Rodrıguez-Herrera et al. (*PRL 104, 253601, 2010*) demonstrate that the SOI effect is extremely sensitive to the distance between the light-focused point and NP, where a small displacement of $\lambda/3$ can generate significant difference in the S3 scattering patterns. In experiment, the authors employed the linearly-polarized laser beam to well control the focused point size and the distance with NP. In contrast, the SOI effect reported in our work works under the excitation of linearly-polarized plane wave, which is the excitation condition in all simulations. In experiment, it is difficult to collect the scattering patterns under the excitation of linearly-polarized plane wave, and the dark-field scattering system used in this work employed the hollow Gaussian beam to excite NPs. However, the light source is the incoherent light source from a tungsten lamp. In this case, the focused point is very large, and can cover lots of NPs, where the four-piece scattering pattern with the same direction is observed from all NPs (see **Fig. S22** for details). Thus, the SOI effect reported in our work is robust, and independent of the distance between the optical axis of Gaussian beam and NPs (**Fig. S22**).

In summary, our finding includes a new effect in Mie scattering and one microscopic electromagnetic dynamics model based on the continuity equation of SAM and OAM, and provides a new perspective to review all of the polarization phenomenon in Mie region.

2. The authors claim to provide a universal theory for the scattering photonic SHE, suggesting it is superior to previous studies in describing this phenomenon. However, the theoretical discussion in the current manuscript is highly technical and relatively qualitative, which weakens the connection to the experimental results and does not clearly demonstrate superiority in explaining the observed photonic SHE. Specifically, with the theoretical discussions, several points remain unclear to the audience such as:

a) The origin of the strong wavelength dependence of the photonic SHE effect.

b) How different plasmonic materials and sizes exhibit varying SOI.

c) Why the nanosphere-on-metal system provides strongly enhanced SHE. A better connection between the theoretical analysis and the experimental results would significantly improve the quality of the manuscript.

> Thanks for this comment. We have significantly improved the electromagnetic dynamics model through introducing the OAM continuity equation and the multipole decomposition theory. With this OAM continuity equation, we get the complete microscopic electromagnetic dynamics model, and clearly reveal the whole transformation process between SAM, OAM, force and polarity in both near- and far-fields. This model bridges the near- and far-field SOI processes, and applies to all of the SOI effects within the framework of Maxwell's theory, such as the Goos-Hänchen Shift, Imbert-Fedorov shift, Magnus effects, T-spin, and optical Skeyrmion.

To connect the theoretical model and experimental results, we simplify the complex polarity distribution in NP into the superposition of multiple dipoles, and then the angular shift between LCP and RCP scattering patterns can be extracted by calculating the intensity and phase of the equivalent dipoles (see **Figs. 4, S18-19** for details). For a system contained only one electric dipole along x axis (p_x) and one magnetic dipole along y axis (m_y/c), an equal dipolar moment with a phase difference of $\pm\pi/2$ is preferred to generated large angular shift (**Figs. S18-19**). Thus, effectively exciting high-order dipoles, or making phase difference between different dipole components is beneficial for enlarging this angular shift.

In this case, the wavelength dependence comes from the intensity and phase of the equivalent dipoles excited in NP, which originates from the polarity retardation effect, and highly depends on the NP size, wavelength, morphology and materials. In fact, this is a Mie scattering. For NP placed on gold film, the mirror charge in gold film can combine with the charge in NP to form strong magnetic dipoles (**Figs. 4c**), leading to significant SOI phenomenon.

In summary, we have construct one complete microscopic electromagnetic dynamics model to reveal transformation process between SAM, OAM, force and polarity, and also the multipole superposition theory to connect the theoretical analysis and the experimental results.

3. The technical accuracy of the current manuscript needs improvement. There are some inconsistencies and inaccuracies, such as:

a. The S3 values shown in Figure 1b are very small, on the order of 10^-16. It appears that the authors did not normalize its value with S0.

b. The definition of S3 (S3 = 2(Ilcp − Ircp)/S0) differs from the conventional definition, including an extra factor of two. This definition would result in a perfect LCP or RCP light having an S3 value of ±2, instead of unity. However, the plot in Figure 4b uses a color bar ranging from -1 to 1, which is potentially misleading.

c. The "green channel" is defined as 480-600 nm in the main text, yet it is stated to be 500-550 nm in the caption of Figure 4.

> Thanks for these comments. We have rewritten the whole manuscript, and corrected all of the mentioned issues.

Referee 2:

1. Limited Originality in Core Findings. The primary claim of this paper is the giant IF effect induced by NSs. However, this phenomenon has been previously documented in multiple publications, such as Ref. [19]. In Ref. [19], the authors theoretically and experimentally demonstrated that spin-orbit interaction (SOI) leads to spin-dependent scattering features in both near-field and far-field regimes. For instance, when x-polarized light illuminates NSs (as investigated here), the left-circularly polarized (LCP) and right-circularly polarized (RCP) components naturally exhibit opposite twisted trajectories, generating the four-piece pattern in the S3 component described by the authors. Additionally, while the authors note that the giant IF is more pronounced for metallic NSs, this observation—like those in prior works—lacks deep mechanistic explanations of the underlying physics.

> Thanks for this comment. As discussed above, the SOI phenomenon reported in this work is a new and universal effect in Mie scattering, and always accompanies the Mie scattering process. This phenomenon is one important property of Mie scattering that has not been reported in other works. Considering the extensive influence of Mie scattering, this SOI phenomenon will also find broad applications in diverse fields, including atmospheric science, remote sensing, biomedical optics, nanophotonics, environmental monitoring, astronomy, materials characterization, and beyonds. Thus, we revise the whole manuscript, including the name, to highlight the Mie scattering nature.

In addition, we construct one complete microscopic electromagnetic dynamics model to clearly reveal the whole SOI process in Mie scattering for the first time. Based on the rigorously-derived continuity equation of photonic spin and orbital angular momenta in an inhomogeneous medium, we clearly display the excitation process with polarized light, the intense SOI in near-field, and the polarized emission in far-field, where the spin momentum, orbit momentum, and optical force are involved. Obviously, this model can apply to all of the SOI phenomenon within the framework of Maxwell's theory, and greatly promote people's understanding of SOI effects.

The key difference with prior works is that the SOI phenomenon reported in this work is one fundamental property of Mie scattering, which is universal and own important influence in Mie region. In sharp contrast, the prior works, however, can't be summarized as one universal property of Mie scattering. For examples,

Deng Pan et al. (*PRL 117, 166803, 2016*) report that the SAM in excitation field can transfer to the twist streamlines near NP (i.e., orbital momentum), which, however, localizes in the near-field, and will vanish when the distance from NP is larger than several wavelengths. This twist streamline in near-field can't propagate to a long-distance unless the wave-vector matching condition is satisfied, for example, putting the NP on a substrate or gold nanowire, where a directional transmission can be observed [**]. Thus, this is a near-field and localized SOI effect, and is inherently different with the SOI effect in Mie scattering;

D. Haefner et al. (*PRL 102, 123903 (2009)*) report a far-field SOI effect when exciting NP with circularly polarized lights, where the far-field Poynting vector S will make an angle with the line of direct sight. But the authors also emphasized that this phenomenon will vanish under the excitation of linearly polarized lights, which is just the case in this work;

O. Rodrıguez-Herrera et al. c demonstrate that the SOI effect is extremely sensitive to the distance between the light-focused point and NP, where a small displacement of $\lambda/3$ can generate significant difference in the S3 scattering patterns. In experiment, the authors employed the linearly-polarized laser beam to well control the focused point size and the distance with NP. In contrast, the SOI effect reported in our work works under the excitation of linearly-polarized plane wave, which is the excitation condition in all simulations. In experiment, it is difficult to collect the scattering patterns under the excitation of linearly-polarized plane wave, and the dark-field scattering system used in this work employed the hollow Gaussian beam to excite NPs. However, the light source is the incoherent light source from a tungsten lamp. In this case, the focused point is very large, and can cover lots of NPs, where the four-piece scattering pattern with the same direction is observed from all NPs (see **Fig. S22** for details). Thus, the SOI effect reported in our work is robust, and independent of the distance between the optical axis of Gaussian beam and NPs (see **Fig. S22** for details).

In summary, our finding includes a new effect in Mie scattering and one microscopic electromagnetic dynamics model based on the continuity equation of SAM and OAM, and provides a new perspective to review all of the polarization phenomenon in Mie region.

2. Incremental Advance in Fundamental Analysis. The authors' detailed discussions on the transient evolution of spin moments, spin moment flux, and spin torques around NSs are valuable for deepening fundamental understanding. However, these analyses lack breakthrough insights. For example:

- It is inherently expected that "their sum gives a force which cancels" due to the structural and incident light symmetries.
- The statement "the ideal dipole only works under the size mismatching condition" is trivial, as it merely restates the dependence on phase retardation scale.
- The assertion "These features explain the origin of the photonic SHE from a NS" is circular, as "these features" are consequences—not causes—of PSHE. The actual origin lies in the chiral characteristics of TM surface wave local optical fields on metals (related to propagation direction) and momentum matching enabled by nanosphere structures via Fourier spectrum analysis.

> Thanks for this constructive comment. We have revised the whole manuscript strictly, and corrected all descriptions of less rigorous.

We note that the reviewer pointed out a very interesting direction to explain the SOI effect in Mie scattering through analyzing the TM surface wave and the momentum matching. This is a very valuable explanation, and we tried to complete this theoretical framework, which includes

i. TM surface wave that drives the polarity to rotate elliptically and make spin fields (analysis based on Mie scattering theory);
ii. The k-space that reveals the near-field spin momentum of each plane-wave component and links the near-field Fourier spectrum and far-field emission (based on the vectorial angular spectrum analysis);
iii. The far-field polarization image that reveals the relationship between the far-field SOI effect and the near-field wave-vectors.

From this theoretical framework, we can get the relationship between the far-field Stokes parameters and the near-field K-space spectrum. This is valuable and under analysis now.

In contrast, the microscopic electromagnetic dynamics model based on the continuity equation of SAM and OAM combines the SAM, OAM, optical force, polarity, and the conversion between them to reveal the complete SOI process in Mie scattering, including the polarity induced by SAM and OAM, the conversion between optical force, SAM and OAM, the rotation of polarity driven by optical force, and the finally formed polarization emission through depolarization and pushing force on NP that is counteracted by surrounding objects. The whole process is more intuitive to readers, and the model based on the continuity equation of SAM and OAM is new. Thus, we improve this SOI model in the up-date manuscript, and will introduce the model based on the TM surface wave and the momentum matching to the next work.

3. Specific Technical and Presentational Issues.
- Fig. 1b: The notation “IF angular shift of 20° ±1° deg” likely contains a typographical error; “±1°” and “deg” are redundant, and the uncertainty may instead be “20° ±2°”.
- The claim “where the wavevector matching condition is not satisfied” is unclear without contextual clarification of the theoretical framework or experimental conditions.
- The description “The charge distribution mostly resembles a magnetic dipole, but sporadically also as an electric dipole” lacks precise physical language; “sporadically” is ambiguous in this context.
- The statement “they cannot account for the phenomenon observed in this work (19, 31)” fails to clarify how the current observations diverge from prior studies or what new physics is claimed.

> Thanks for these comments. We have rewritten the whole manuscript, including all of the issues mentioned above.

Referee 3:

1. Photonic spin-orbit interactions can be manifested in different ways. The system under study, consisting of an isolated nanosphere under the excitation of a linearly polarized plane wave, is not quite aligned with the common configuration of IF shifts, where the incoming light hitting on the interface is a circularly polarized beam. If the plane wave is considered as an approximation of a beam in the paraxial limit, then the angular shift vanishes (see Ref. 4). Similarly, for the nanosphere lattices, diffraction should not be confused with the IF effect.

> Thanks for this kind reminder. We note that the common configuration of IF shifts usually refer to a circularly-polarized Gaussian beam hitting on an interface. Thus, we remove this term, and employ a more appropriate description (SOI effect) in the update manuscript.

2. Although not having the same geometry, I feel the present result may share some spirits with the unidirectional excitation of guided waves reported in Science 340, 328-330 (2013). It is actually not surprising that light scattered into different directions carries different polarization states. In the extreme case, a certain polarization can be redirected by a scatterer to one direction, analogous to the IF effect.

> Thanks for this comment. As described in the up-date manuscript, the polarization emission phenomenon in this work is a universal SOI effect in Mie scattering, which is one kind of scattering phenomenon in far-field, and doesn't require specific illumination conditions, geometrical morphology, structure, or others. In other words, the previously-reported polarization scattering usually require specific conditions, such as the non-paraxial conditions[6,24], specifically-designed interfaces[10,25-27], and wavevector matching in spin-locked coupling via evanescent near-fields[7,20]. While the SOI effect reported here is more fundamental effect, which may become one building block (or one basic principle) for designing the target polarization emission.

It should be noted that the unidirectional excitation of guided waves reported in Ref. (*Science 340, 328-330 (2013)*) is indeed a near-field phenomenon, where the SAM in excitation field is converted to the orbit momentum in near-field. This orbit momentum is usually one property of the excited surface wave, and owns a wave-vector ***k*** that depends on the SAM in excitation field, which can lead to a unidirectional propagating phenomenon when the wave-vector matching condition is satisfied. While the SOI effect in our work is a far-field Mie scattering phenomenon.

In fact, this is an accident invention in one experiment to measure the LCP and RCP scattering patterns from chiral NPs. We found that different chiral NPs own similar S3 pattern (i.e., the four-leaf pattern of the same rotation angle), and this pattern is surprisingly almost independent on the chirality of NPs. Then, we made systematical investigation on this phenomenon and found that it is a universal phenomenon in all Mie particles of different morphologies, including normal NPs, ultra-spherical NPs, nano cubes and nano decahedra. These results indicate that this is a new SOI effect in Mie scattering.

3. In Figs. 1 and S4, it appears that the effect in discussion is most prominent in the ultraviolet region and decreases dramatically at longer wavelengths. I wonder how the dielectric property of the nanospheres affects the magnitude of the effect. In Fig. 1b, at 200 nm wavelength gold is no longer a plasmonic material. In Fig. 1c, the sphere size and scattering direction are both changed, which cannot support the argument “The IF angular shift varies with the wavelength” on Page 3. In Fig. S7, the values of refractive index are not consistent in the figure and in the caption.

> Thanks for this comment. As described in the up-date manuscript, the observed SOI effect is one fundamental properties of Mie scattering, and highly depends on the excited equivalent dipoles. For the perfect nano sphere in Mie region, the excited magnetic dipoles in nano sphere is key for enlarging the angular shift between LCP and RCP scattering patterns (see **Figs. S18-19** for details), where an equal intensity and a phase difference of 45 deg between the electric ($\mathrm{p_x}$) and magnetic dipoles ($\mathrm{m_y/c}$) will generate a maximum angular shift of 90 deg. In other words, the angular shift depends on the relative relationship between the excited dipoles, rather than the intensity of one or more certain dipoles. Thus, the angular shift doesn't depend on the resonance properties of NPs.

We also investigate the ideal case with the Mie theory through changing the refractive index for a NP of fixed size. As shown in **Figs. S5-6**, the angular shift vanishes for $\mathrm{n} = 0$ or $\kappa = 0$, and increases when n or κ increases, where n and κ are the real and imaging parts of refractive index, respectively. **Figs. 1b-c** show the scattering patterns from the same NP with size of 80 nm, where the wavelength is 200 nm and 800 nm, respectively. The issues in figures, including **Fig. S7**, have been corrected.

4. In the same paragraph, the authors claim “these phenomena are further verified by analytical solutions from Mie theory….” I don’t see how Mie theory is applied to the analysis. But given the sphere sizes and wavelength ranges, I do agree that Mie theory is a very helpful tool to unveil the mechanism and verify whether the present effect has a non-resonant nature. To this end, the paragraph discussing the failure of an ideal dipole model does not seem to give much insight. With different implementations of Mie theory (e.g. Nanoscale 8, 10441-10452 (2016), J Chem. Phys. 155, 224110 (2021), etc.), it should be possible to separate the contributions from each multipole in forming the polarization state in any direction.

> Thanks for this comment. As mentioned above, the observed SOI effect is is one fundamental properties of Mie scattering. The angular shift comes from the elliptically-rotated polarities induced on different locations of NP, which emit polarized lights to different directions. These polarities can be simplified as some equivalent dipoles based on the multipolar decomposition theory[37], and the coherent superposition of these emission field from different dipoles generates the angular shift between LCP and RCP scattering patterns. Thus, we employ the multipolar decomposition theory to analyze the relationship between the structure parameters and the angular shift (**Figs. 4, S18-19**). Mie theory was

used in calculating the scattering patterns to show how the scattering pattern change with the refractive index and wavelength (**Figs. S5-7**).

We note that the works in (*Nanoscale 8, 10441-10452 (2016)*, *J Chem. Phys. 155, 224110 (2021)*) describe a chiral emission from the system combining chiral emitter and Mie particles. This distinguishes significantly from the SOI effect in Mie scattering, where the NPs can be chiral or not, and the scattering patterns is symmetrical in the entire space.

5. The authors are encouraged to discuss the connections between the present results and those reported in their recent paper Nano Lett. 25, 1158-1164 (2025).

> Thanks for the kind reminder. The work (*Nano Lett. 25, 1158-1164 (2025)*) reports one method to enhance the chiroptical effect in scattering from NPs, and also demonstrate that almost all of the synthesized NPs are chiral, which comes from the tiny symmetry broken in the self-assembled process. The key working mechanism is that the metal film placed underneath greatly suppress the out-of-plane dipole, while the tiny symmetry broken in NPs bring phase difference to different dipole components in x or y directions, leading to strong chiroptical effect in the backward scattering field. In contrast, this work discusses the difference between LCP and RCP scattering patterns from Mie particles, where the NPs can be chiral or achiral, and the chiroptical effect is also not necessary. In fact, we tried to combine these two works before, but failed because one refers to the chiral geometry, and the other refers to the SOI, which are two independent phenomena.